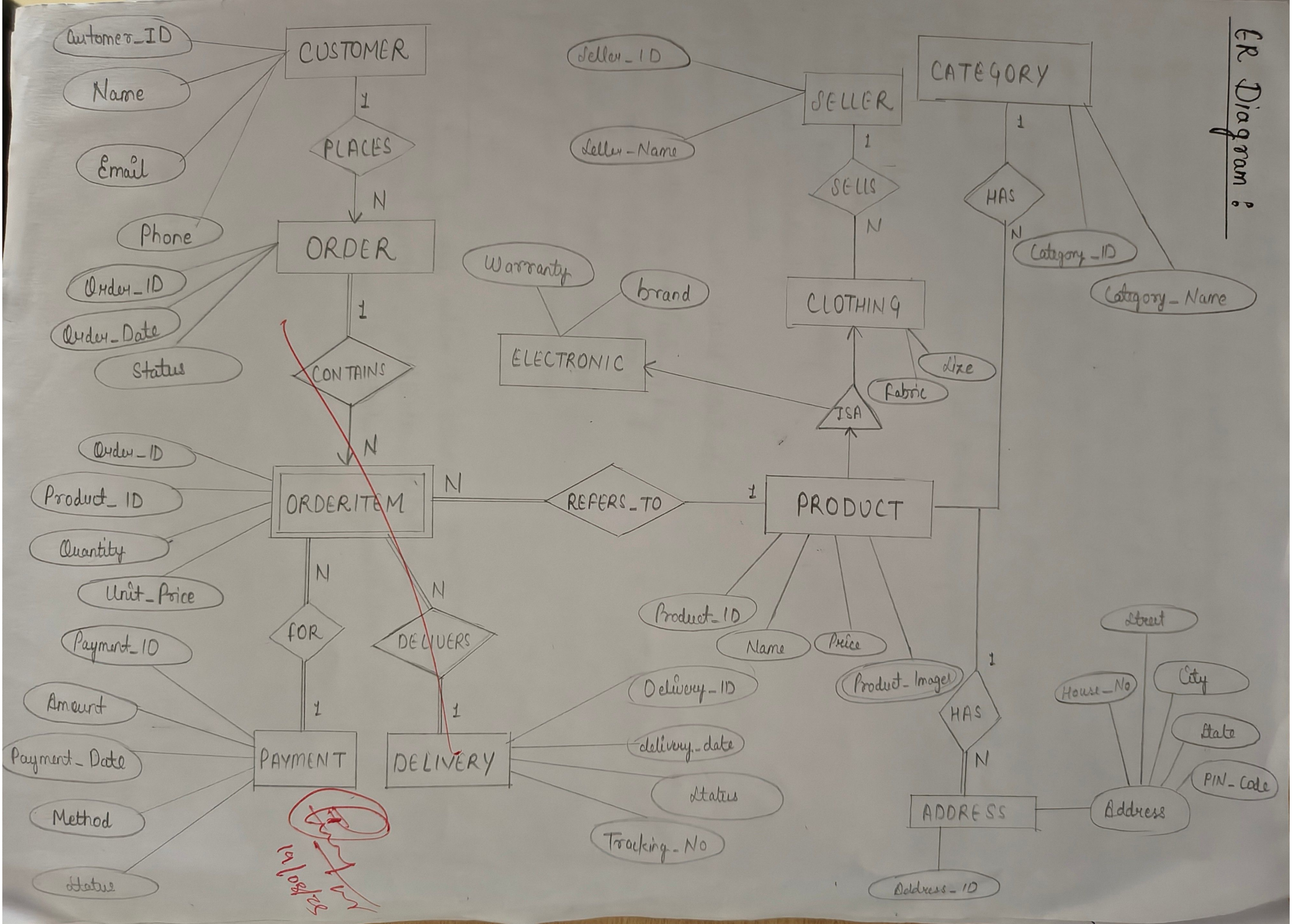
EXPERIMENT No-1

AIM:

To design an entity-Relationship (ER) diagram for an Indian e-commerce platform including customer, Product, Order, Orderitem, Seller, Category, Payment, Delivery and Address while representing Primary keys, composite and multi-valued attributes, weak entities, specialization of Product and participation constraints.


ER Diagram :
CUSTOMER
Customer_ID
Name
Email
Phone
PLACES
1
N
ORDER
Order_ID
Order_Date
Status
CONTAINS
1
N
SELLER
Seller_ID
Seller_Name
SELLS
1
N
CATEGORY
HAS
1
N
Category_ID
Category_Name
CLOTHING
Size
Fabric
ELECTRONIC
Warranty
brand
ISA
ORDERITEM
Order_ID
Product_ID
Quantity
Unit_Price
N
REFERS_TO
1
PRODUCT
Product_ID
Name
Price
Product_Image
FOR
DELIVERS
N
N
1
1
PAYMENT
Payment_ID
Amount
Payment_Date
Method
Status
DELIVERY
Delivery_ID
delivery_date
Status
Tracking_No
HAS
1
N
ADDRESS
Address_ID
Address
Street
City
House_No
State
PIN_Code

Participation Constraints

(1) Customer - Order : Customer has partial participation ; Order has total participation.
(2) Order - Orderitem : Order and Orderitem have total participation.
(3) Order - Item Product : Order item has total participation ; product has partial.
(4) Seller - Product : Product has total participation ; Seller has partial participa
(5) Category - Product : Product has total participation ; Category has partial.
(6) Order - payment : Order has total participation ; payment has total participation.
(7) Order - delivery : Order has total participation ; delivery has total participation.
(8) Customer - Address : Customer has partial participation ; Address has total.

ER Notations Used

(a) Rectangle → entity
(b) double - Rectangle → Weak entity
(c) diamond → relationship
(d) Oval → attribute
(e) double Oval → multi - valued attribute
(f) underlined attribute → Primary key
(g) ISA Triangle → Specialization
(h) double line → total participation
(i) Single line → partial participation

Output

The ER diagram for the Indian E- commerce platform is successfully designed with all specified entities, attributes, relationships, keys, weak entity, multi-valued and Composite attributes, specialization and participation Constraints.